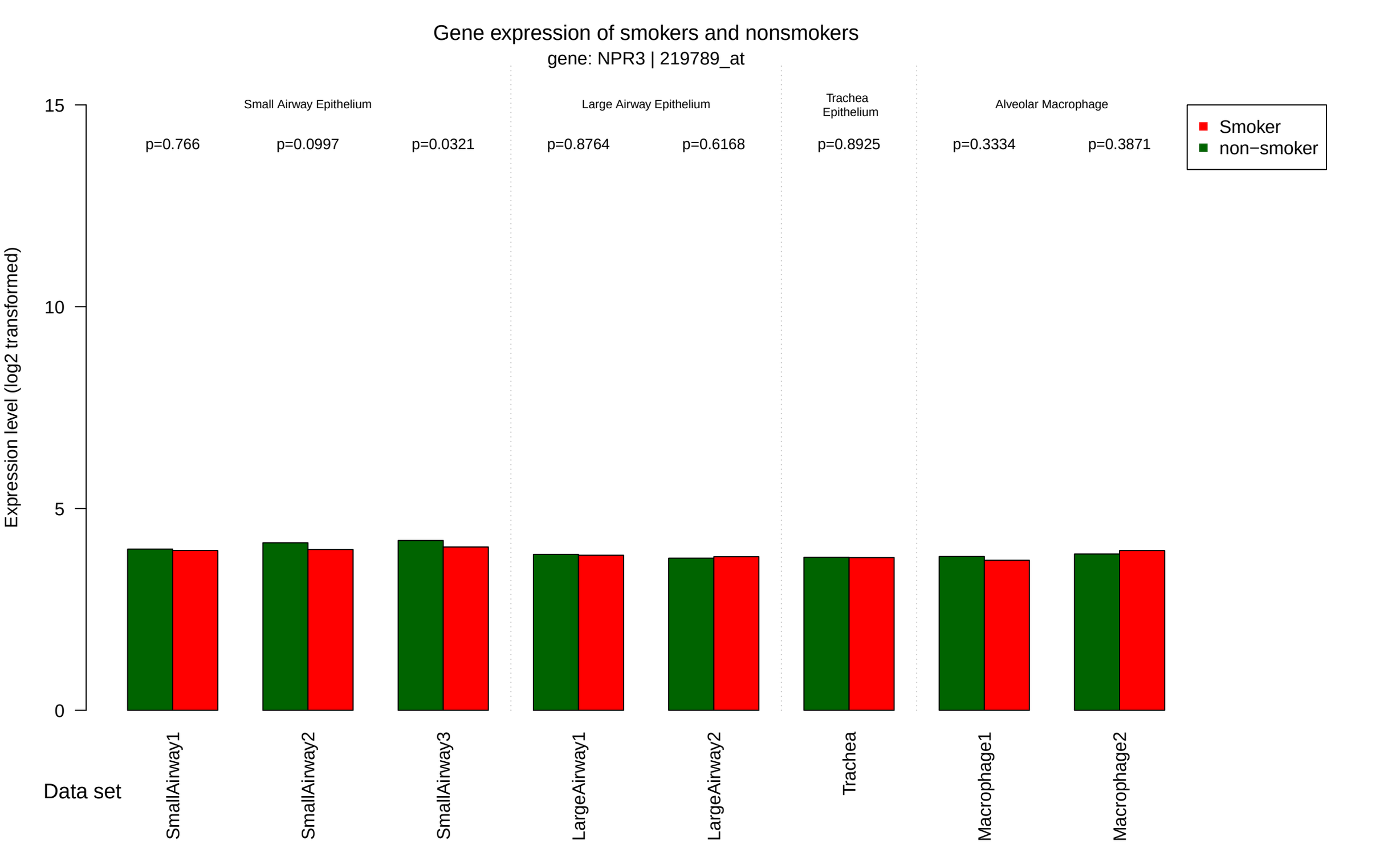
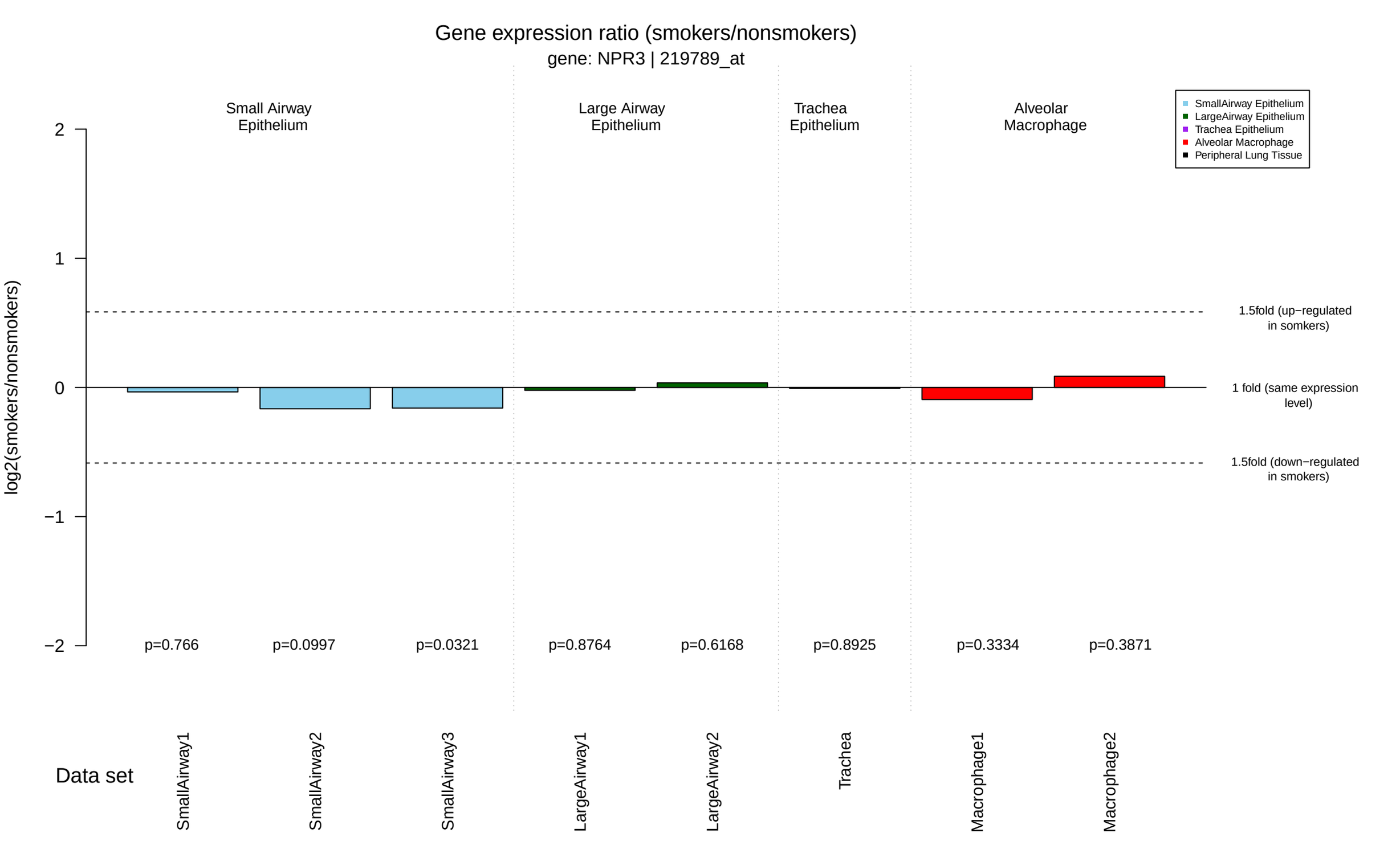
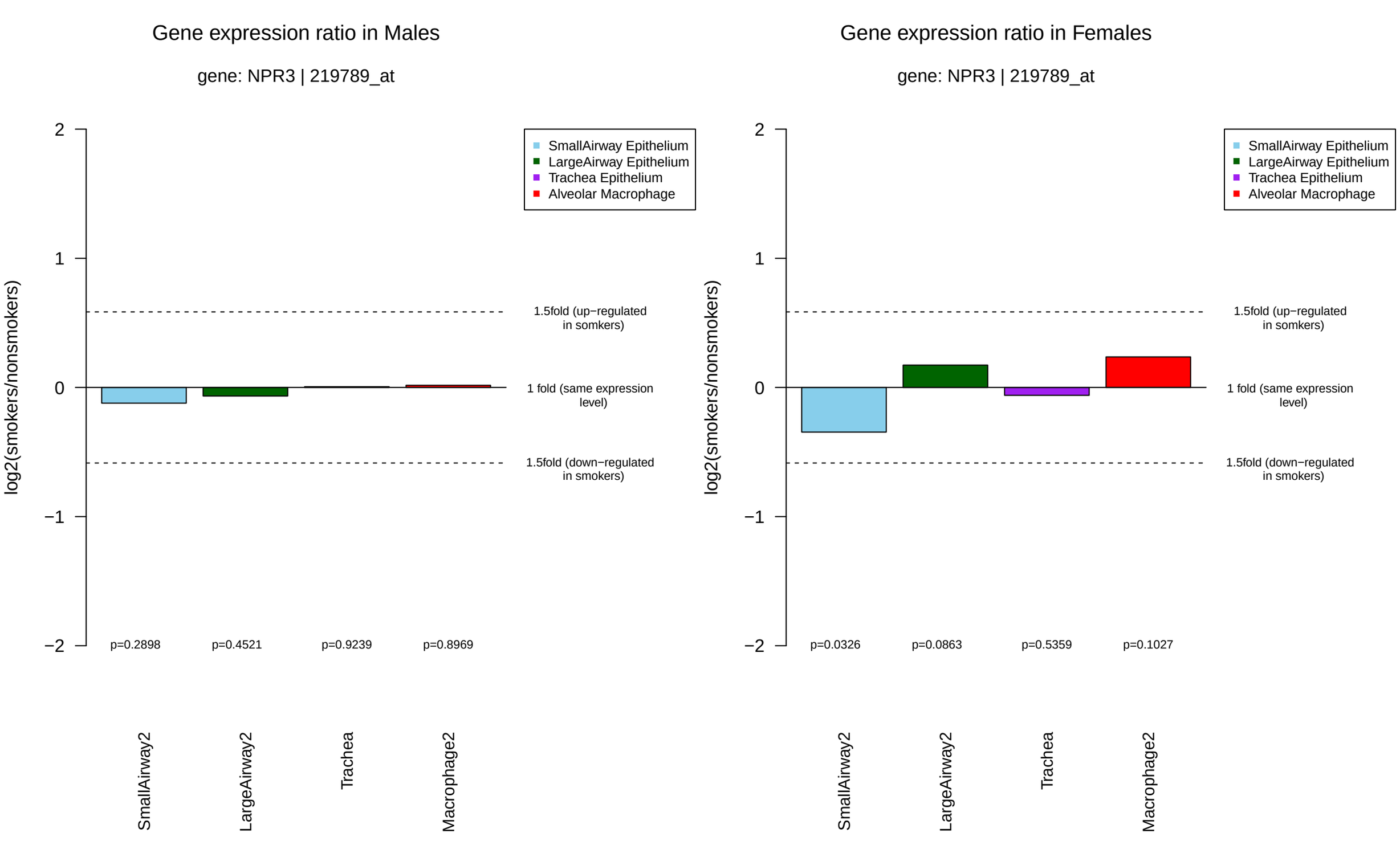
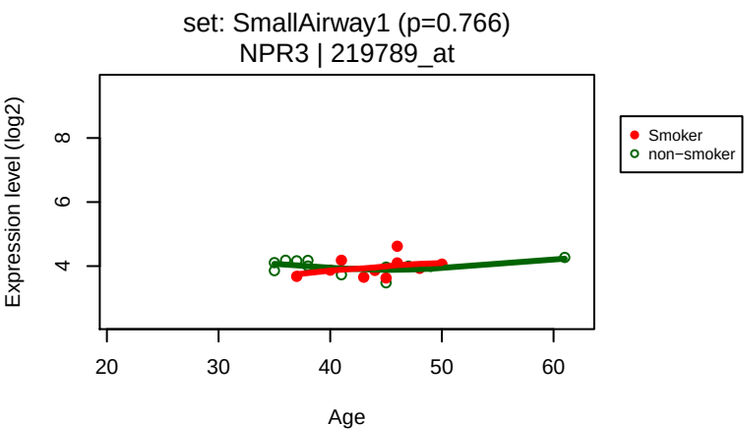
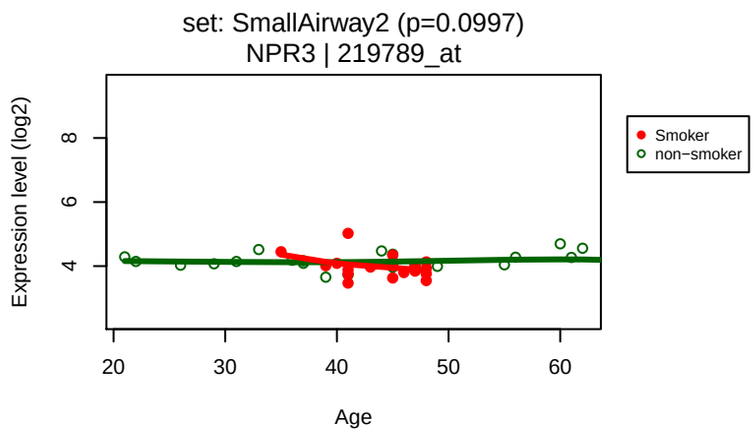
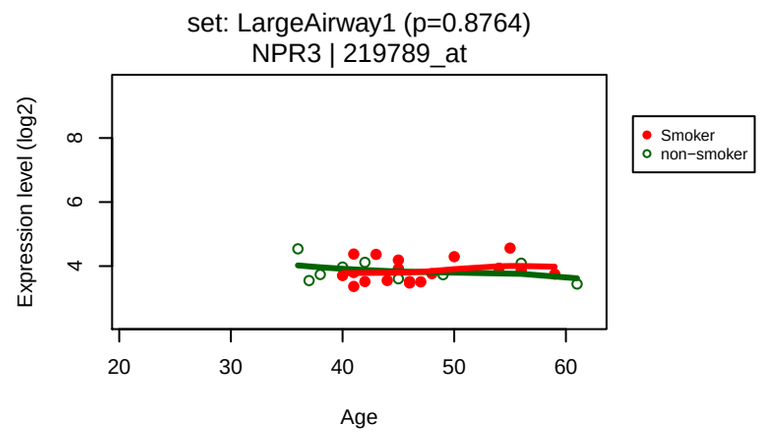

Gene expression of smokers and nonsmokers
gene: NPR3 | 219789_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.766
p=0.0997
p=0.0321
p=0.8764
p=0.6168
p=0.8925
p=0.3334
p=0.3871
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

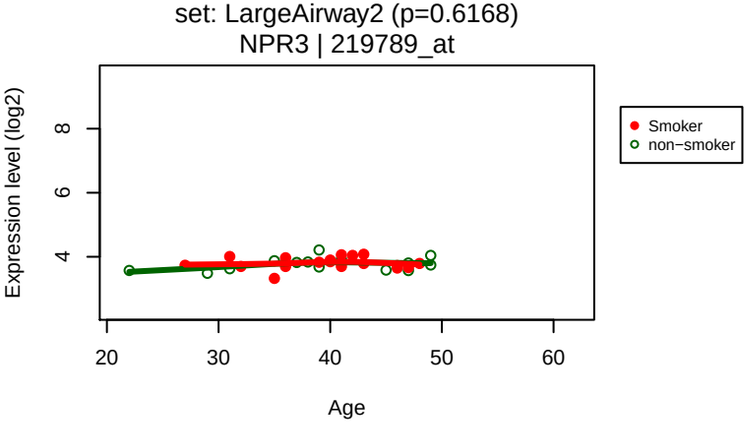

Gene expression ratio (smokers/nonsmokers)
gene: NPR3 | 219789_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.766
p=0.0997
p=0.0321
p=0.8764
p=0.6168
p=0.8925
p=0.3334
p=0.3871
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

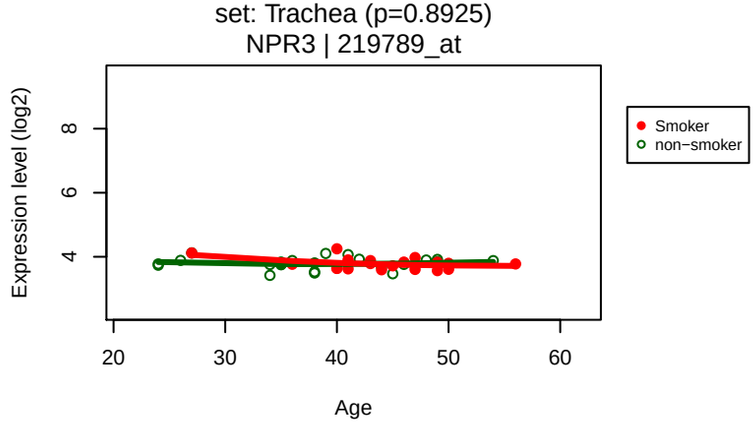

Gene expression ratio in Males
gene: NPR3 | 219789_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2898
p=0.4521
p=0.9239
p=0.8969
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPR3 | 219789_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0326
p=0.0863
p=0.5359
p=0.1027
SmallAirway2
LargeAirway2
Trachea
Macrophage2

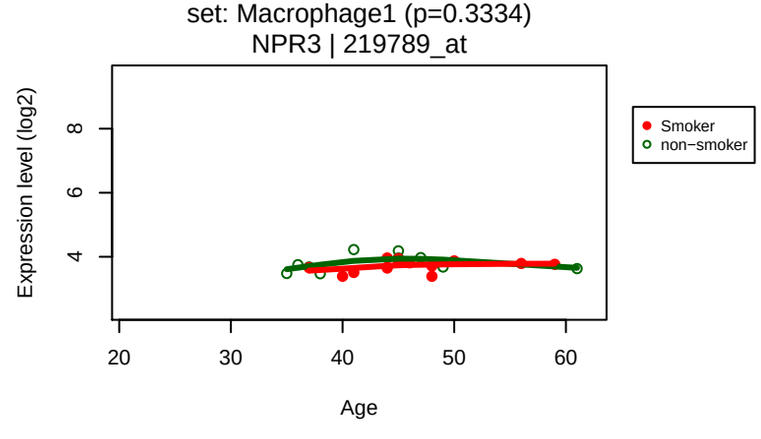

set: SmallAirway1 (p=0.766)
NPR3 | 219789_at
Expression level (log2)
Age
Smoker
non−smoker

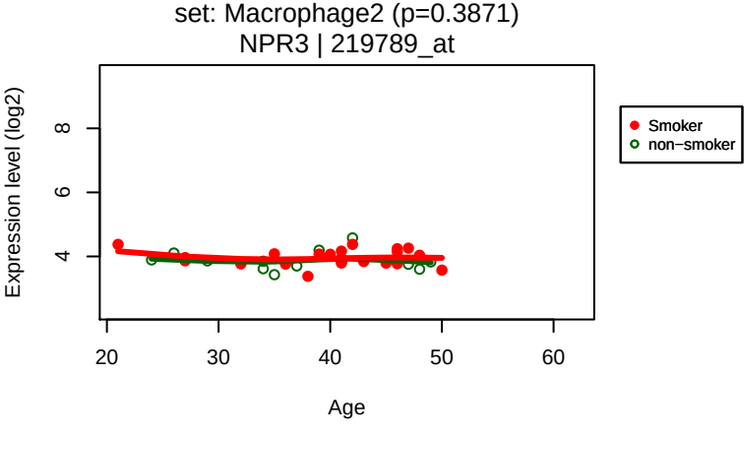

set: SmallAirway2 (p=0.0997)
NPR3 | 219789_at
Expression level (log2)
Age
Smoker
non−smoker

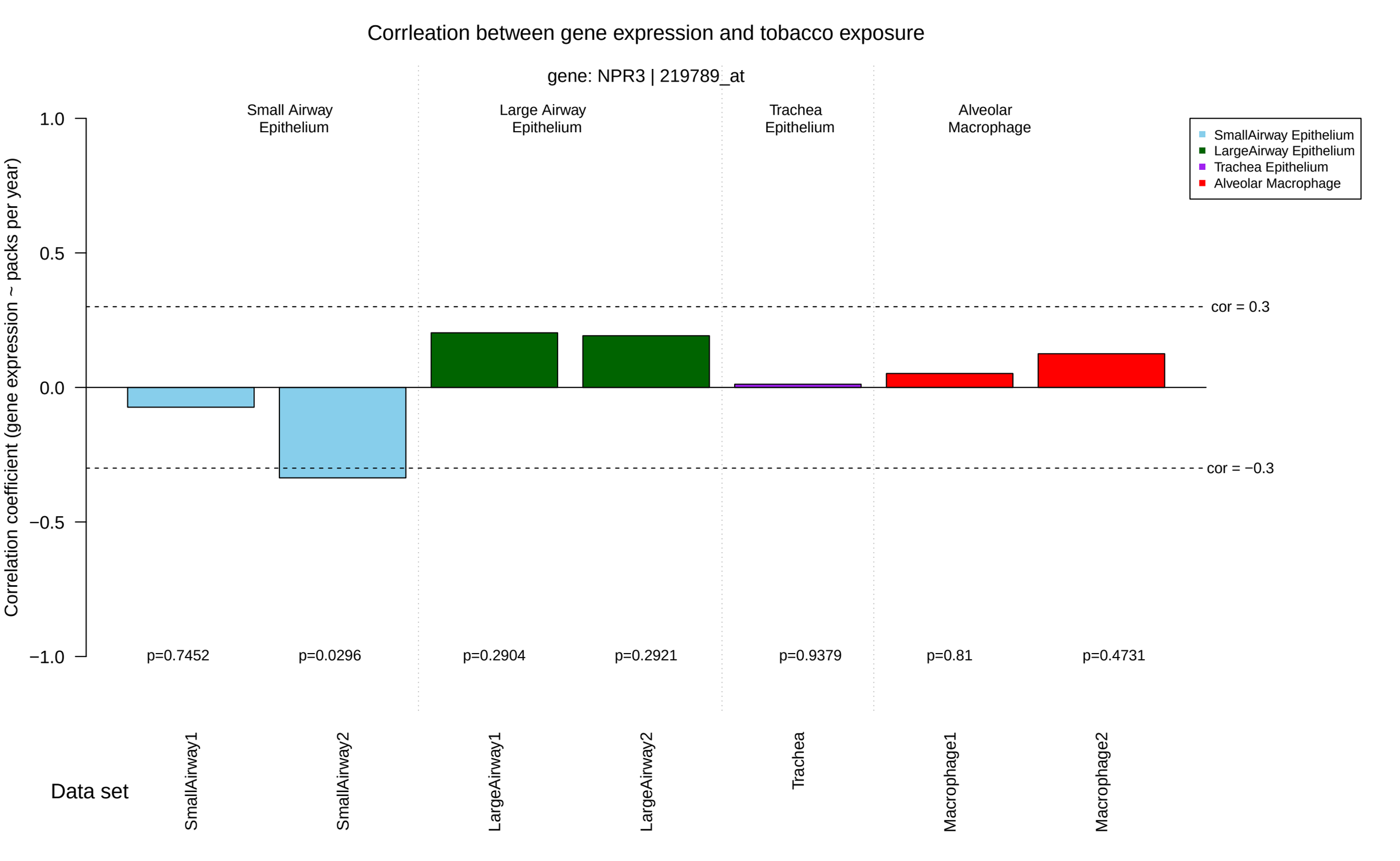

set: LargeAirway1 (p=0.8764)
NPR3 | 219789_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6168)
NPR3 | 219789_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8925)
NPR3 | 219789_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3334)
NPR3 | 219789_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3871)
NPR3 | 219789_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPR3 | 219789_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7452
p=0.0296
p=0.2904
p=0.2921
p=0.9379
p=0.81
p=0.4731
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2